Majorana Modes in the Machine

Week 8: NISQ Reality Check – Noise Model and Failure Modes

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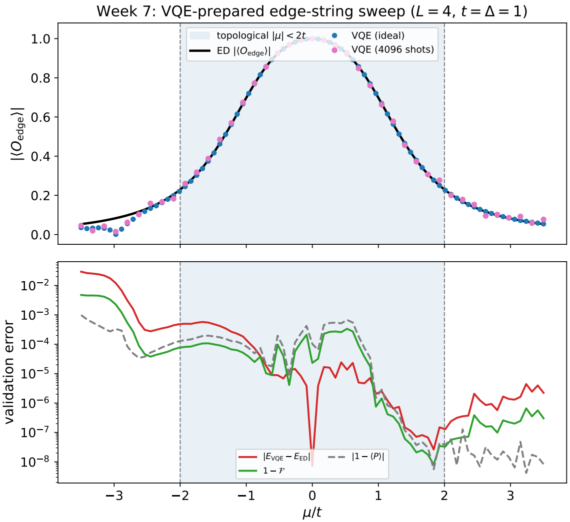
What Week 7 Gives Us



Noiseless baseline

- Prepared by parity-constrained VQE
- Edge string measured ideally and with shots
- ED validates energy, parity, fidelity, string value

What Noise Can Break

State preparation

- VQE ansatz gates no longer implement exactly $U(\theta)$.
- Entangling-gate errors reduce edge coherence.
- Relaxation can bias the parity sector.

Measurement

- Readout flips corrupt the bitstring parity product.
- Shot noise broadens the string estimator.
- A small true signal becomes hard to distinguish from zero.

Main risk

Noise can flatten the edge-string contrast between $|\mu| < 2t$ and $|\mu| > 2t$, even when the noiseless circuit is correct.

Noise Channels to Add

Channel	**Physical meaning**	**Expected effect**
Depolarizing error	Random Pauli error after gates	Shrinks correlations and string contrast
Readout error	Classical bit flip during measurement	Biases ± 1 parity-product estimator
Amplitude damping	T_1 relaxation toward $\|0\rangle$	Can change particle number and parity balance
Phase damping	T_2 dephasing	Destroys coherent edge-to-edge correlations

Two Experiments to Keep Separate

Frozen-parameter noise test

- Use $\theta^*(\mu)$ from noiseless Week 7.
- Add noise only during circuit execution.
- Measures readout and circuit degradation at fixed preparation.

Noisy optimization test

- Optimize the VQE cost under noisy estimates.
- Tests whether the optimizer can still find good parameters.
- Harder and more expensive; should come second.

Engineering rule

First isolate measurement/circuit degradation; only then study optimizer degradation.

Circuit-Level Noise Protocol: Formalism

Objective: Formalize the mapping from the noiseless VQE sweep to the noisy density matrix.

For the Week 7 optimized parameters $\theta^*(\mu)$, the ideal pure state is $\rho_{\text{id}} = |\psi(\theta^*)\rangle\langle\psi(\theta^*)|$. We define the physical execution via completely positive trace-preserving (CPTP) maps:

$$\rho_{\text{noisy}} = \mathcal{E}_{\text{readout}} \circ \mathcal{E}_{\text{meas-basis}} \circ \mathcal{E}_{\text{gate}}(\rho_{\text{id}}) \tag{1}$$

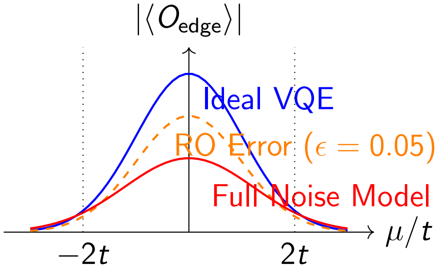
- **Gate Noise ($\mathcal{E}_{\text{gate}}$):** Modeled via local depolarizing channels after each 2-qubit gate. $\mathcal{D}_p(\rho) = (1-p)\rho + \frac{p}{4^2-1}\sum_{i\neq 0} P_i \rho P_i$.
- **Basis Rotation ($\mathcal{E}_{\text{meas-basis}}$):** Ideal unitary mapping for $X_0 Z_1 Z_2 X_3$.
- **Readout Assignment ($\mathcal{E}_{\text{readout}}$):** Asymmetric bit-flip channel $p(0|1)$ and $p(1|0)$ applied to the diagonal elements of the density matrix.

Frozen-Parameter Test: Readout vs. Depolarizing Channels

Protocol: Evaluate $\mathrm{Tr}(\hat{O}_{\text{edge}}\rho_{\text{noisy}})$ using frozen $\theta^*(\mu)$ from the ED/ideal VQE baseline.

Expected Behavior:

- The ideal ED baseline peaks at $\langle O_{\text{edge}}\rangle = 1.0$ at $\mu = 0$.
- *Readout error alone* linearly scales the entire curve by $(1-2\epsilon)^4$, preserving the functional shape.
- *Gate noise* broadens the variance $\mathrm{Tr}(\rho^2) < 1$, flattening the topological contrast disproportionately near the critical point $\mu_c = \pm 2t$.



Phase-Boundary Failure Criterion

We must define mathematically when the topological transition at $|\mu| = 2t$ becomes indistinguishable from the trivial vacuum due to the combined noise channels.

Failure Criterion: The diagnostic is considered to have failed when the topological signal inside the bulk ($|\mu| < 2t$) cannot be resolved from the statistical and hardware noise floor with a 3σ confidence interval.

Specifically, failure occurs if the maximum observable contrast satisfies:

$$\max_{\mu} \left| \langle \hat{O}_{\mathrm{edge}}(\mu) \rangle_{\mathrm{meas}} \right| \leq \frac{3}{\sqrt{N_{\mathrm{shots}}}} + \delta_{\mathrm{SPAM}} \tag{2}$$

where δ_{SPAM} is the systematic bias inferred from the parity drift observable, $|1 - \langle \hat{P} \rangle_{\mathrm{noisy}}|$.

Next Week

Next milestone

After the frozen-parameter tests, run noisy VQE optimization and compare raw versus mitigated readout.